

# Volume 18 Swarm Intelligence - Swarm Intelligence Part 4

Version v94 draft

README.md

## Swarm Intelligence Part 4

Version: v94 draft

Status: Draft

Document ID: MAL-V18-V94-README

### Purpose

This release folder contains the official v94 documentation set for Swarm Intelligence Part 4.

### Scope

- Covers Collaboration.
- Covers Swarm Memory.

### Acceptance

- All documents contain implementation-level guidance.
- The release PDF and ZIP are generated and verified.
- MANIFEST and RELEASE\_NOTES are updated for v94.

## Swarm Intelligence Part 4 Index

- [guide-18-94-01-collaboration.md](#) - Collaboration
- [guide-18-94-02-swarm-memory.md](#) - Swarm Memory

# Collaboration

Title: Collaboration

Version: v94 draft

Status: Draft

Document ID: MAL-V18-COLLABORATION-01

## Purpose

Provide official implementation guidance for Collaboration inside Mariam AI Enterprise OS.

## Scope

- Defines the recommended workflow, required evidence, roles, inputs, outputs, interfaces, and acceptance checks.
- Applies to architecture, engineering, operations, testing, governance, and release work connected to this topic.
- Excludes temporary experiments unless they are promoted into official documentation.

## Responsibilities

- Owners maintain the document, examples, and acceptance evidence.
- Implementers follow the workflow and record deviations.
- Reviewers verify security, quality, traceability, and operational readiness.
- Operators confirm deployment, monitoring, recovery, and maintenance requirements where relevant.

## Operating Model

- Plan the work from documented requirements and trace it to Blueprint or Specifications.
- Implement in small increments with tests, observability, and rollback behavior.
- Review changes against standards, threat model, failure modes, and acceptance evidence.
- Release only when documentation, tests, and operational notes are current.

## Inputs

- Architecture Library documents, implementation tickets, configuration, schemas, policies, runtime context, and stakeholder constraints.
- Prior release notes, known risks, incident learnings, test results, and acceptance criteria.

## Outputs

- Updated documentation, implementation plans, review evidence, test reports, operational runbooks, and release artifacts.
- Traceable decisions that explain why the selected approach is correct for Mariam.

## Interfaces

- Git repository, CI/CD pipeline, runtime services, event bus, storage, dashboards, issue tracker, and release packaging.

- Human interfaces: review boards, chief office, department leads, operators, testers, and governance approvers.

## Security

- Apply least privilege, secret isolation, signed releases where applicable, and audit-ready evidence.
- Do not expose tenant data, credentials, customer materials, private model artifacts, or privileged runtime outputs.
- Escalate security-sensitive deviations before implementation or release.

## Metrics

- Lead time, review time, test pass rate, defect leakage, incident count, recovery time, documentation freshness, and acceptance completion.
- Topic-specific service metrics must be linked to dashboards and release notes.

## Testing

- Verify expected behavior, failure behavior, recovery behavior, and evidence generation.
- Include automated tests where implementation exists and manual acceptance checks where documentation is the deliverable.
- Record test gaps explicitly in release notes.

## Acceptance Criteria

- The workflow is clear enough for a new team member to execute without oral context.
- Required inputs, outputs, owners, interfaces, tests, and evidence are documented.
- Risks and security considerations are visible before release.
- The document is linked from INDEX, TRACEABILITY, ACCEPTANCE, MANIFEST, and release notes.

## Traceability

| Source                     | Relationship                     |
|----------------------------|----------------------------------|
| MAL-V18-COLLABORATION-01   | Governs Collaboration.           |
| Volume 05                  | Engineering standards reference. |
| RELEASE_NOTES_v94_draft.md | Release evidence.                |

## Version History

| Version   | Date       | Status | Notes                                           |
|-----------|------------|--------|-------------------------------------------------|
| v94 draft | 2026-06-29 | Draft  | Added official documentation for Collaboration. |

# Swarm Memory

Title: Swarm Memory

Version: v94 draft

Status: Draft

Document ID: MAL-V18-SWARM-MEMORY-02

## Purpose

Provide official implementation guidance for Swarm Memory inside Mariam AI Enterprise OS.

## Scope

- Defines the recommended workflow, required evidence, roles, inputs, outputs, interfaces, and acceptance checks.
- Applies to architecture, engineering, operations, testing, governance, and release work connected to this topic.
- Excludes temporary experiments unless they are promoted into official documentation.

## Responsibilities

- Owners maintain the document, examples, and acceptance evidence.
- Implementers follow the workflow and record deviations.
- Reviewers verify security, quality, traceability, and operational readiness.
- Operators confirm deployment, monitoring, recovery, and maintenance requirements where relevant.

## Operating Model

- Plan the work from documented requirements and trace it to Blueprint or Specifications.
- Implement in small increments with tests, observability, and rollback behavior.
- Review changes against standards, threat model, failure modes, and acceptance evidence.
- Release only when documentation, tests, and operational notes are current.

## Inputs

- Architecture Library documents, implementation tickets, configuration, schemas, policies, runtime context, and stakeholder constraints.
- Prior release notes, known risks, incident learnings, test results, and acceptance criteria.

## Outputs

- Updated documentation, implementation plans, review evidence, test reports, operational runbooks, and release artifacts.
- Traceable decisions that explain why the selected approach is correct for Mariam.

## Interfaces

- Git repository, CI/CD pipeline, runtime services, event bus, storage, dashboards, issue tracker, and release packaging.

- Human interfaces: review boards, chief office, department leads, operators, testers, and governance approvers.

## Security

- Apply least privilege, secret isolation, signed releases where applicable, and audit-ready evidence.
- Do not expose tenant data, credentials, customer materials, private model artifacts, or privileged runtime outputs.
- Escalate security-sensitive deviations before implementation or release.

## Metrics

- Lead time, review time, test pass rate, defect leakage, incident count, recovery time, documentation freshness, and acceptance completion.
- Topic-specific service metrics must be linked to dashboards and release notes.

## Testing

- Verify expected behavior, failure behavior, recovery behavior, and evidence generation.
- Include automated tests where implementation exists and manual acceptance checks where documentation is the deliverable.
- Record test gaps explicitly in release notes.

## Acceptance Criteria

- The workflow is clear enough for a new team member to execute without oral context.
- Required inputs, outputs, owners, interfaces, tests, and evidence are documented.
- Risks and security considerations are visible before release.
- The document is linked from INDEX, TRACEABILITY, ACCEPTANCE, MANIFEST, and release notes.

## Traceability

| Source                     | Relationship                     |
|----------------------------|----------------------------------|
| MAL-V18-SWARM-MEMORY-02    | Governs Swarm Memory.            |
| Volume 05                  | Engineering standards reference. |
| RELEASE_NOTES_v94_draft.md | Release evidence.                |

## Version History

| Version   | Date       | Status | Notes                                          |
|-----------|------------|--------|------------------------------------------------|
| v94 draft | 2026-06-29 | Draft  | Added official documentation for Swarm Memory. |